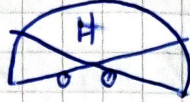
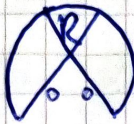


LG

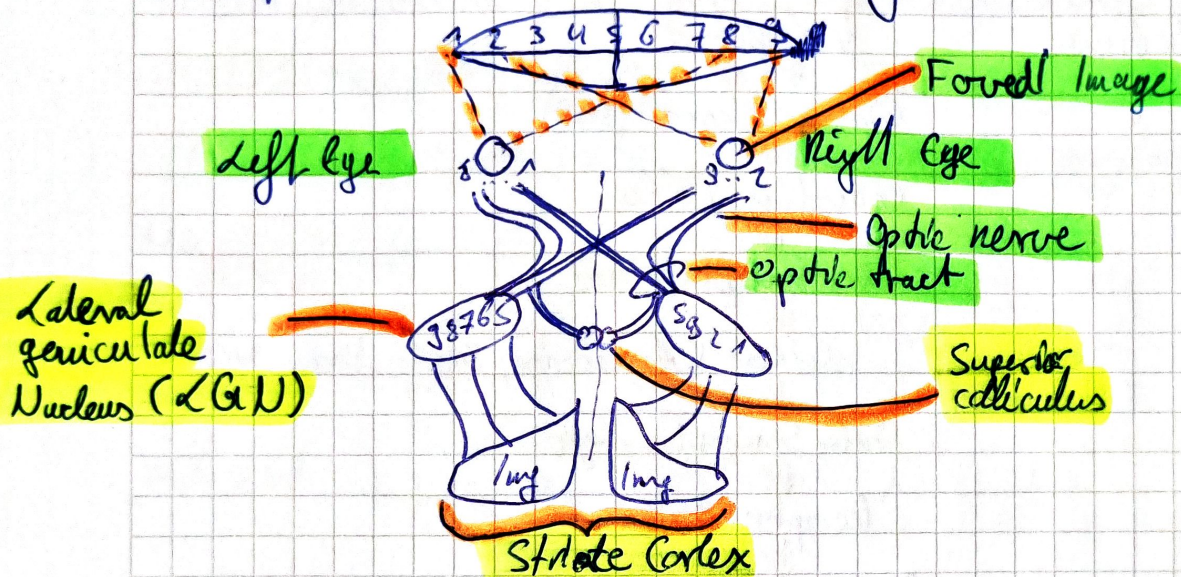
Bio-inspired Hierarchical Vision

- Different visual fields of view in nature

eg: rabbit & human $\Rightarrow$ human brain more complex $\Rightarrow$ integrates info from ^{both} eyes



- Bio-inspired Visual Information Processing



- Human Visual Areas

$\rightarrow$ Visual information processing in occipital lobe (back of brain)

$\rightarrow$ Different areas for different visual information

$\rightarrow$ V1: Edge detection

$\rightarrow$ V2: More complex patterns, parts of figures

$\rightarrow$ V3: Colour, Motion

$\rightarrow$ (V4: Orientation)

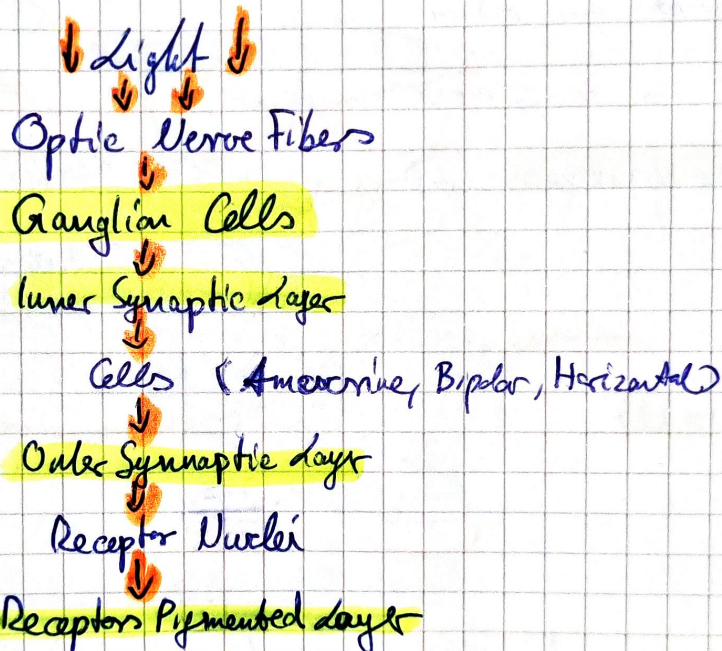
$\rightarrow$ V5: Motion, Eye Movements

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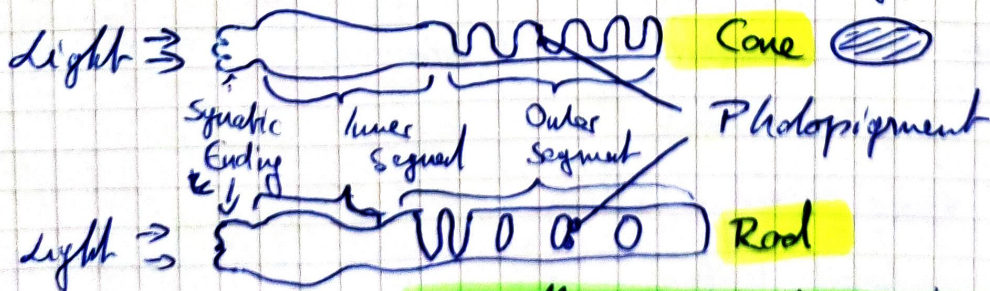
Retinal Processing

- Stimuli from retina project to four subcortical regions
- Lateral Geniculate Nucleus (LGN): provides input to V1 (Edge Detection)
- Hypothalamus: Controls circadian cycle (body clock)
- Pretectum: controls pupillary light reflex
- Superior Colliculus: controls eye movements

The Retina



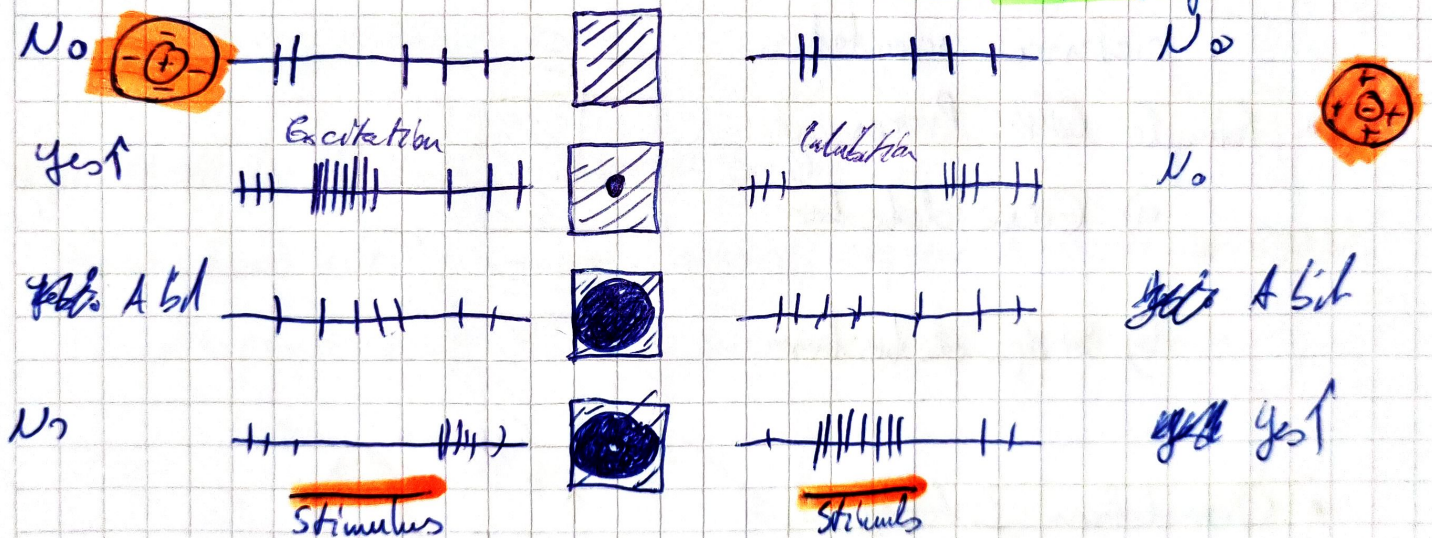
- Retina: inside the roof and the cone cells
color vision; not fovea; they function best in bright light



rod cells are almost entirely responsible for night; outer edge of retina

• Receptive Fields

- Region in which the presence of stimulus will alter the firing of that neuron
- Light stimulus evokes action potential (AP) in ON-ganglion cells
- Frequency increases with sensory strength
- Derivation of AP determines retinal area: receptive field
- ON: excitatory influence on stimulus, center of RF
- OFF: inhibitory influence on stimulus, periphery of RF



• Visual Area 1 (V1)

→ Cell types:

- Simple Cells: Line/Edge Detector, highly selective
- Complex Cells: Orientation Detector, implements invariant features eg: translational invariance
- Hypercomplex Cells: Angle/Length Detector; composed of several complex cells

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• Hubel and Wiesel

→ Cells in **striate cortex** respond best to **bars of light** rather than to spots of light

Simple cells: bars of light / bars of dark

Complex cells: bars of light & bars of dark

→ **Orientation tuning**:

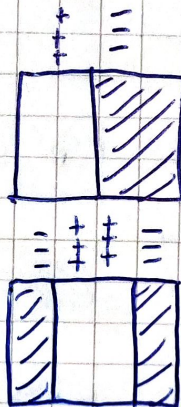
→ Tendency of neurons in **striate cortex** to respond more to bars of **certain orientations** and less to others

→ Response rate falls off with angular difference of bar from preferred orientation

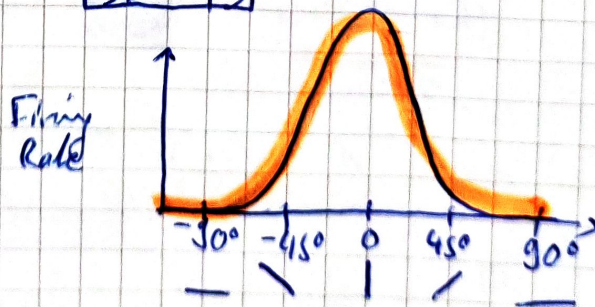
• Simple Cell Processing

a) **Edge detector**

b) **Stripe detector**



• Orientation Detection



• Ventral and Dorsal Streams

Dorsal: **Visual Spatial Location** - Where?

Ventral: **Object Recognition** - What?



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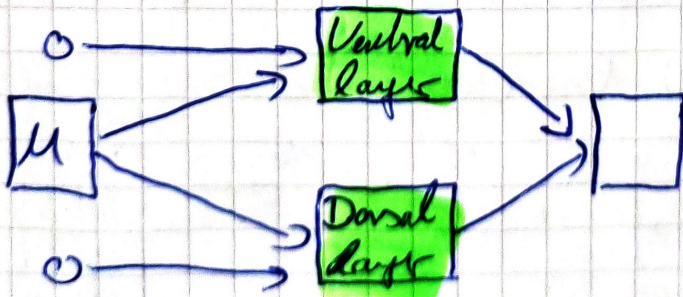
Higher Visual Processing

- Dichotomy of object processing in occipital lobe
- Partitioning found by Ungerleider
- Separation underpinned by lesion studies
- Ventral Stream: projects to temporal cortex
 - Neural encoding of shape, colour, texture, ... (what)
- Dorsal Stream: projects to parietal cortex
 - Neural encoding of objects spatial information (where)
 - also sometimes referred to as how-path due to sensory-motor transformation

eg: NN for Ventral & Dorsal Stream

How about storing multiple sensory sequences in an RNN?

Can the sensory info still be encoded in two streams?

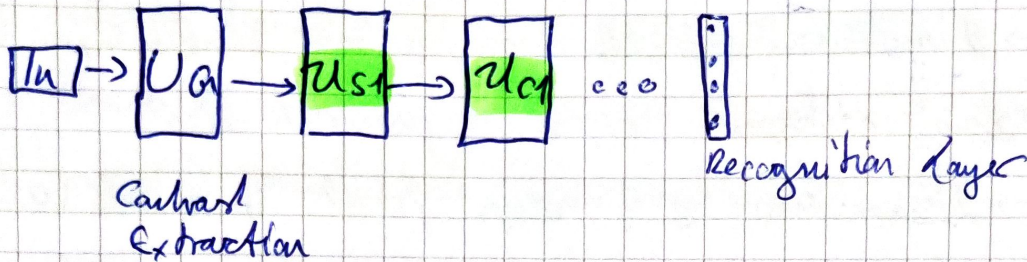


• Hierarchical Vision Models: Neocognition (Fukushima)

→ Hierarchical Multilayer NN for visual recognition

→ Alternate planes of simple S-cells (feature extraction) and complex C-cells (positional errors)

→ Resemble processing stages in visual cortex

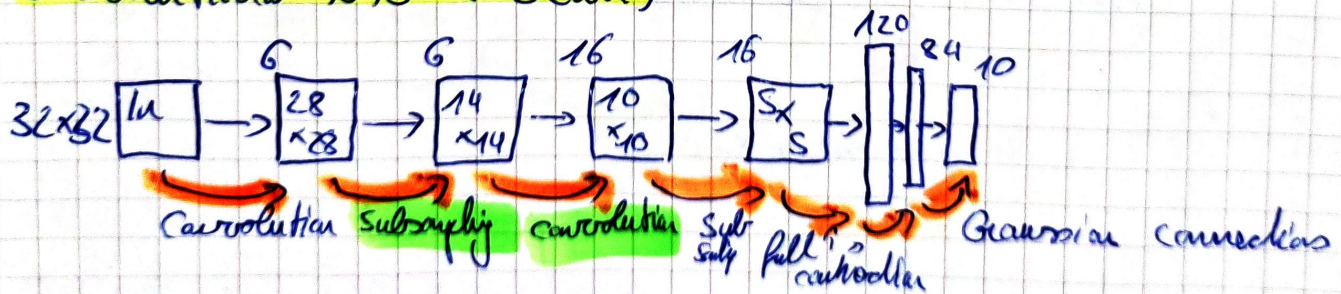


→ S-cells trained to particular feature present in receptive field

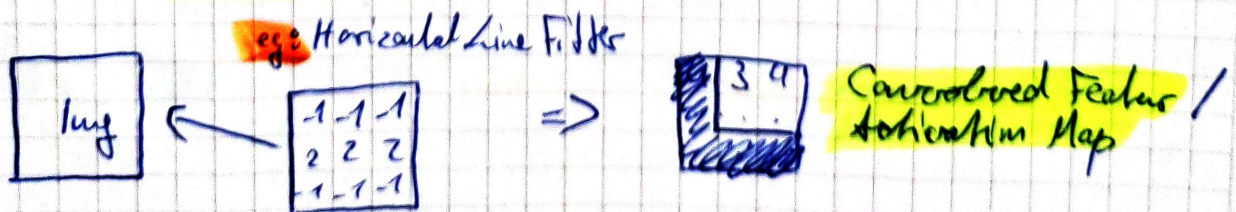
→ C-cells inserted to correct for positional errors: receive responses from S-cells coding for the same feature

→ Training of S-cells with unsupervised or supervised methods; only S-cells have learning inputs

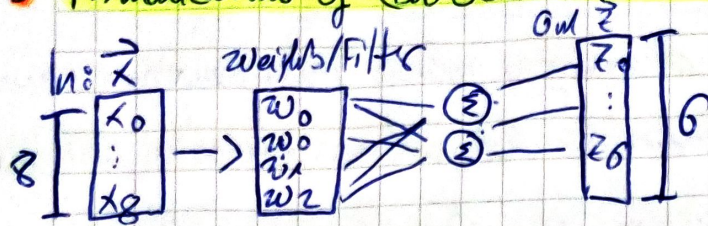
• Convolutional NN (LeCun)



• Convolutional Kernels as Filters



Formalization of Convolutional Layer (1D)



Receptive Field Size $r = 3$

Output Dot product (convolution)

$$z_i = \sum_{j=0}^{r-1} w_j x_{i+j}, \quad i \in [0, N]$$

Stride: how much you jump $s = 1$

Number of patches $N = \frac{|X| - r}{s} + 1$

eg: $\begin{bmatrix} 2 & -1 \\ 2 & -2 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = z_i$

→ Zero Padding p : amount of elements added to the input vector to obtain an output vector of the same size as the input (i.e. $N = N$)

eg: $\begin{bmatrix} 2 & -1 \\ 2 & -2 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = z_i$

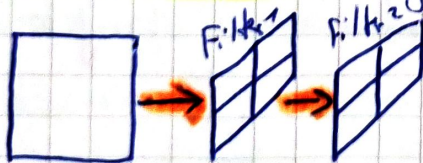
Convolution with Multiple Filters

Each filter creates an activation map.

Activation of i -th neuron on k -th activation map

$$z_{i,k} = \sum_{j=0}^{r-1} w_{j,k} x_{i+j}$$

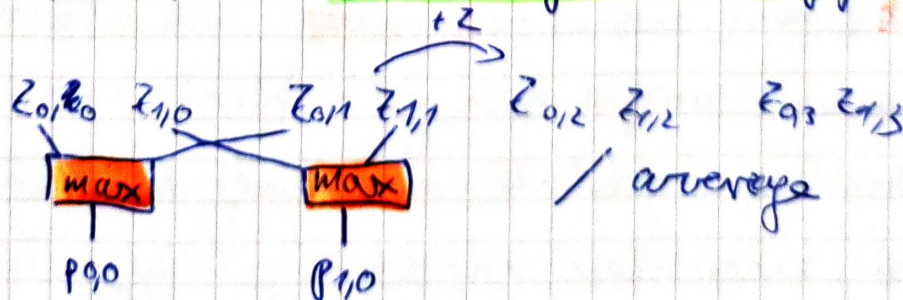
2D Convolutional Layers with multiple Filters



Dimension of output $n_f \times r$ for n_f filters.

Dimensionality reduction by pooling!

Pooling



Pooling layer $\vec{p}$ with receptive field $r_p = 2$

and stride size $s_p = 2$

$$\dim(\vec{p}) = 4 \times 2, \quad \dim(\vec{z}) = 8 \times 2$$

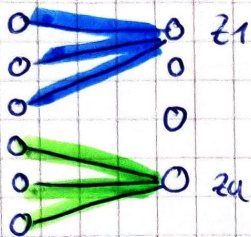
L6

Biologically Plausible Convolutional NN

to detect common local features across an image by applying the same weight

- CNN require weight sharing, which real neurons cannot do
- Locally connected networks do not share weights but perform worse than CNNs in image classification tasks

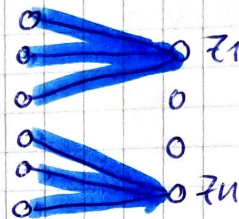
locally connected



$$w_1 \neq w_4$$

Convolutional

⇒ same kernel



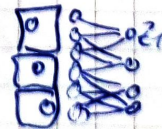
$$w_1 = w_4$$

- How to bridge the gap between the biologically plausible locally connected networks and the well-performing but less plausible CNN?

- Two Mechanisms:
 - Data augmentation via multiple image translation
 - Dynamic weight sharing

Data augmentation:

- Done by showing a locally connected network multiple translations of the same image ⇒ simultaneously!
- This makes neurons in a channel react similarly to the same input
- Requires longer training times
- only small performance improve

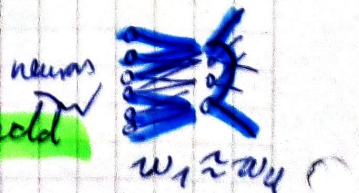


$$w_1 \approx w_2$$

$$w_3 \approx w_4$$

Dynamic weight sharing:

- Done by adding lateral connectivity to a locally connected network and allowing learning via Hebbian plasticity
- This helps equalize the weights of the laterally connected



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$$\begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} \} \Rightarrow \begin{matrix} \cdot \\ \cdot \\ \cdot \end{matrix}$$

• Biologically Plausible CNN

→ Dynamic weight sharing

→ Lateral connections are added between every k^{th} neuron in the input and output layer

→ A sleep phase is introduced every training iteration where the weights of the laterally connected neurons i are updated by Hebbian rule:

$$\Delta w_i \propto - \left(z_i - \frac{1}{N} \sum_{j=1}^N z_j \right) x - \delta(w_i - w_i^{\text{init}})$$

sleep phase
↑
init

Stochastic
→ Gradient Descent on the sum $(z_i - z_j)^2$

• Local Interpretable Model - Agnostic Explanations (LIME)

→ Explain classifier locally, in the neighborhood of a sample

→ Create local dataset by perturbing the sample & dropping interpretable units

→ Classify these perturbed samples

→ Weight the perturbations by their similarity to the sample

→ Train an interpretable classifier with this perturbed data

• Summary

→ Visual info processed in different brain areas

→ Images projected from retina to striate and extrastriate cortex, filtered according to different features

→ Visual Processing inspired hierarchical computational models:

→ Object recognition (Fukushima, LeCun)

→ Deep Learning Models